HACK APERTUS LIEBEFELD · RESEARCH & EXPERTISE NAVIGATOR CHALLENGE

BearMiner

Dig Bern gold from life-science partners.

A Research & Expertise Navigator that turns scattered university web pages into source-backed experts, units and partners — powered by Apertus.

BUILT FOR Bern Economic Development Agency

TEMPTEAM · OCTOBER 2026



THE PROBLEM

To find an expert today, you have to know **where** to look before you know **what** to look for.

"glucose sensors" | continuous glucose monitoring | CGM | biosensing | metabolic monitoring | wearable sensing

3

source ecosystems: UniBE subdomains, ARTORG, Diabetes Center Berne

5+

page types per institute: faculty, group, person, project, service

0

places a company can ask one question and get traceable answers

OUR SOLUTION

Two sides of the same search, meeting on evidence.

RESEARCH SIDE

What Bern can do

- Focused crawl of all three ecosystems
- Up to 500 ranked research highlights
- EN/DE medical keywords, verbatim evidence, source URL



MATCH

COMPANY SIDE

Who Bern should call

- Reads a pitch PDF, website or text, any language
- Builds a verified medical profile
- Collaboration potential 0–100 and P1–P4 priority

HOW IT WORKS

Public web in, source-backed answers out.

1

Collect

Inventory hosts only; robots.txt and rate limits respected

RULES

2

Select

Findings, projects, technologies, services, each with a quote

RULES

3

Understand

Need → terms and EN/DE synonyms; company medical profile

APERTUS

4

Retrieve & score

BM25 over records; five-part, recomputable match rate

RULES

5

Explain

Keeps relevant units and writes why each one matches

APERTUS

Apertus v1.5 — language understanding

Deterministic code — crawling, verification, scoring

COMPANY SIDE · OUR LOGIC

From any pitch to a Bern match score.

1	**Read** Pitch PDF, website or text, any company	RULES
2	**Translate** Any language to English, e.g. Japanese sources	APERTUS
3	**Medical profile** Indications and needs, each quoted from the source	APERTUS
4	**Match** Themes vs Bern capabilities and crawled records	RULES
5	**Score** Six capped parts, 0–100, checkable by hand	RULES
6	**Hand off** P1–P4 queue with the Bern units to call first	RULES

MATCH SCORE = SIX PARTS, MAX 100



25+
a match

50+
gold match · P2

70+
very strong · P1

Not life science? Score 0, never a made-up partial match.

WHAT A RESULT LOOKS LIKE

Every answer carries its own proof.

DIABETES CENTER BERNE · PROJECT

The Moonwalk Project

glucose · hypoglycemia

> “After the end of the study, algorithms will be developed to build a warning system for nocturnal hypoglycemia…”

dcberne.com/en/project/the-moonwalk-project/

Real record from our live crawl test: keywords, verbatim quote, public source.

SAMPLE COMPANY · WEARABLE CGM, ZURICH

85.5



Very strong · P1 — prioritise outreach

Capability overlap	16.3/25
Partnership readiness	17.6/20
Medical focus	13.9/15
Domain fit	14.2/15
Swiss proximity	13.5/15
Signal depth	10/10

Why not higher: “Swiss company but no Canton of Bern site yet.”

PURPOSEFUL AI · TRACEABLE BY DESIGN

Apertus where language is hard. Rules where trust matters.

Apertus does

Bridges vocabulary — “glucose sensors” finds CGM, biosensing, metabolic monitoring

Reads companies — any-language pitch to a medical profile

Explains matches — a short, sourced reason per result

Code does

Decides the score — same input, same number, offline

Checks every claim against the source text

Collects and ranks research without a model or GPU

Quote or drop

No source quote (≥ 85% match), no item

Closed choice

Only candidates it was given, with URLs

Honest gaps

“No reliable match” instead of a guess

Verified location

Bern points never from model memory

TECHNICAL RIGOUR

The score separates real fits from noise.

SAMPLE COMPANY	COINS	SCORE
Glucose-sensor start-up, Burgdorf (BE)		87.0
Wearable CGM company, Zurich		85.5
Surgical navigation, Munich		71.8
Glucose diagnostics, Japanese source		57.3
Medical devices, Japanese, few needs		46.5
Fintech — out of scope	—	0.0

Offline run of our scorer on the repo samples: no LLM, no crawled Bern records loaded.

5 → 7 live medical pages produced seven evidence-backed records with keywords

30 labelled benchmark examples, 10 per source; build fails below 90% precision

EN·DE·JA sources handled — translated before matching

Tests for robots, redirects, budgets, evidence gates and the 500-record cap

SOVEREIGN · AFFORDABLE · EXTENSIBLE

Runs on Bern's terms, air-gapped if needed.

Sovereign

Apertus v1.5 on an on-premise or Swiss-cloud endpoint. SQLite and Python — no foreign closed service at runtime.

Low cost

Crawling and scoring need no GPU. LLM answers are cached, so re-runs are instant and identical.

Extensible

A new institution is one config entry — then re-run crawl and enrich. No pipeline rewrite.

```
# src/sources.json — add a source
{ "id": "newinst", "name": "New Institute", "url": "https://…", "institution": "…" }
```

FROM PROTOTYPE TO PILOT

Give us one quarter and one real company inbox.



A TOOL FOR Bern Economic Development Agency

WEEKS 1–4

Complete crawl

Full run over all three ecosystems on a Swiss-hosted Apertus endpoint.

WEEKS 5–8

Expert review

Agency staff review highlights; feedback tunes rules and synonyms.

WEEKS 9–12

Live outreach

Score incoming company requests and hand off P1 leads to Bern units.